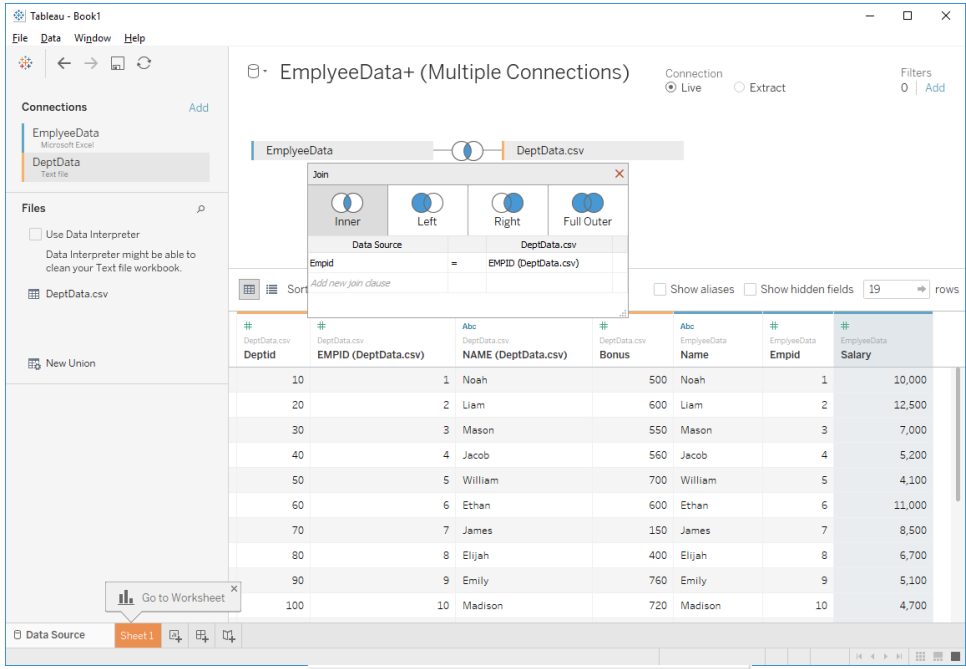
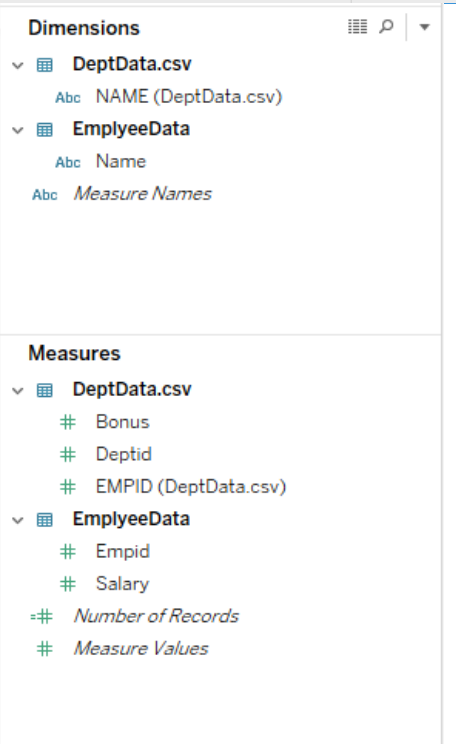
How to combine cross Databases

Cross Database joins

- In cross joins, we combine data from multiple data sources and represent in the form of the graph.
- For this section, we use “DeptData.csv” and “EmplyeeData.xlsx”. The first file is “comma separated file” in notepad and the second file is excel file.
- Sheet1:
  - Open Tableau, on “Connect” select “Microsoft excel” and select “EmplyeeData.xlsx”.
  - Then click on “add” next to “Connection”, choose “Text file” and select “DeptData.csv”.
- Afterwards, the list of fields are shown and their type, their source are also demonstrated.
- There are two “EMPID” and “Name” fields, the one without having its source name next to it is considered as primary.
- There are various joints, here we consider “Inner” join. The “Empid” in both data bases are considered as the join field.
- Sheet1(cntd):
  - Click on “Sheet1”. On the left side, fields in each database are separated and stored in “Dimensions” and “Measures”.



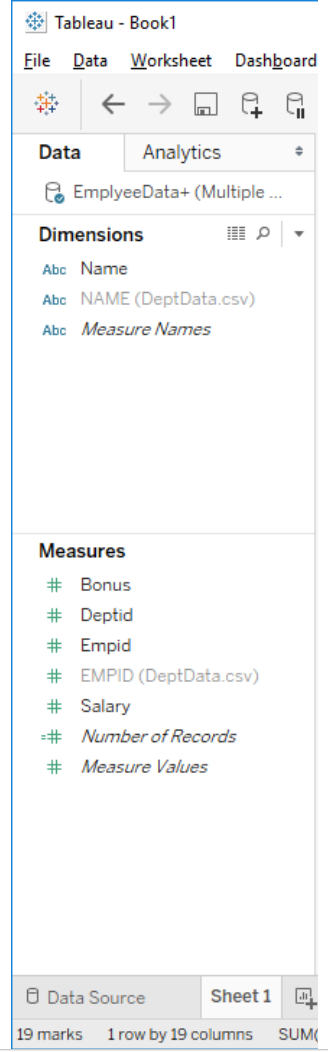


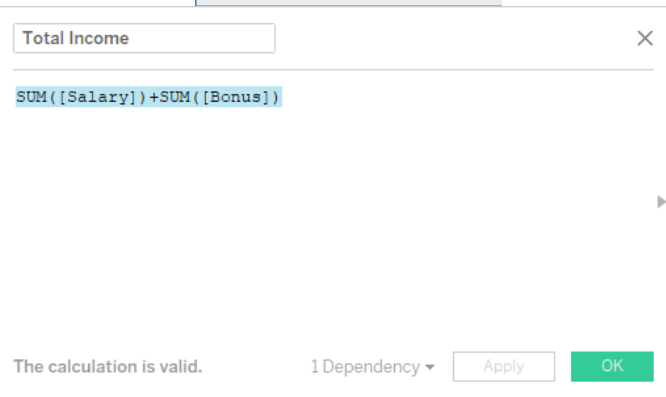
Cross Database joins (Cntd.)

- Sheet1(cntd):
  - Right click on the blank space and select “Group By Folder”, immediately the name of data sets are disappeared, and both the columns are merged into one folder.
  - You can hide “NAME(DeptData.csv)” and “EMPID (DeptData.csv)”, since there are duplicated fields. Each one has primary and secondary fields.
  - Double click “Salary”
  - Double click “Name”
  - Drop “Bonus” into the “Color”
  - Click on “Sum(Bonus)” and sort it
  - Chane the color of the chart from “Blue” to “Orang-Blue Diverging”
  - Gently move your cursor on each bar and figure out the name od people, their salaries and bonuses
  - Write a meaningful caption and title for the graph
  - Export it as a bmp image file

Q1: Create a bmp image file from sheet1 and insert it into your document.
Q2: Is there any logical relationship between salary and bonus? Explain more.

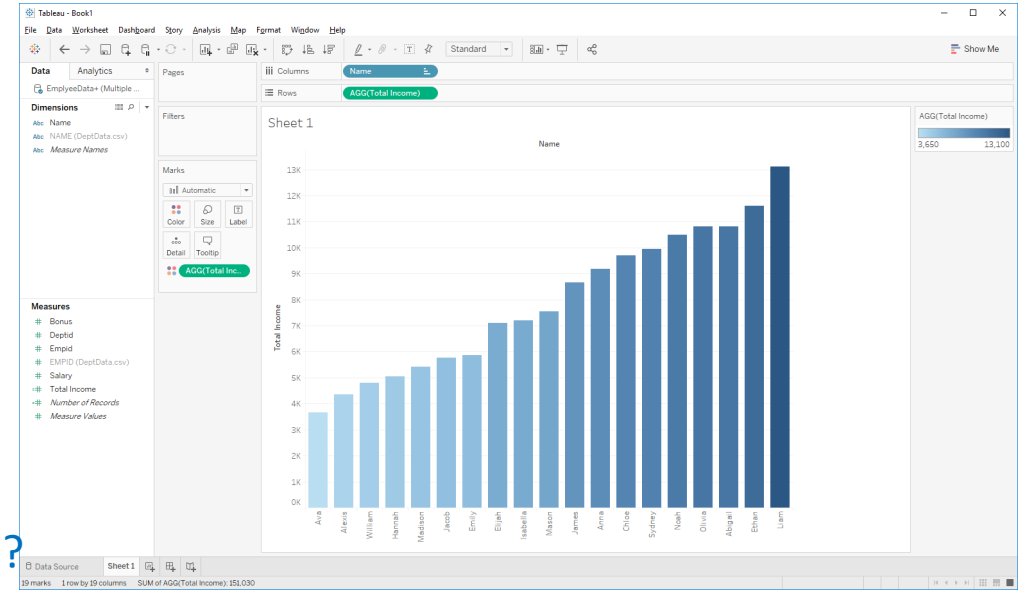
- Clean up the screen





Cross Database joins

- Now we want to have a chart that shows sum of salary and bonus.
- Sheet2:
  - Right click on the blank space on the “Measures” and select “Create Calculated Field”. Name this new field as “Total Income”.
  - Write “SUM([Salary])+SUM([Bonus])” to customize a new field, then Click ok
  - Drop “Name” in “Columns”
  - Drop “Total Income” in “Rows” and “Color”
  - Click on “AGG(Total Income)” and sort it



Q3: Who got the maximum and minimum “Total income” and how much?

- Clean up the screen

Scatter charts

Create an interactive scatter chart

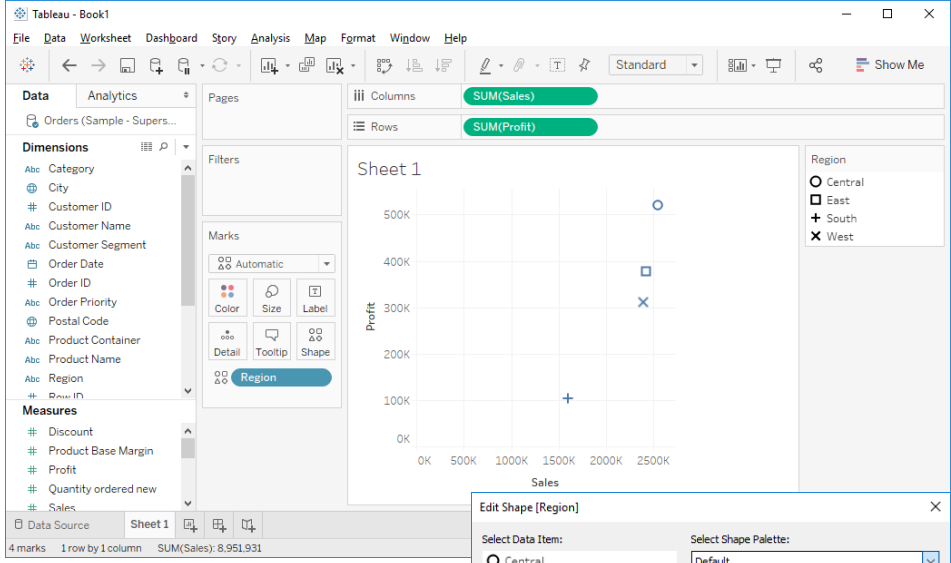
- Interactive scatter charts can be created using filters.
- Open a new workbook. Open “Sample – Superstore.xlsx”, select “Orders”, and click “Sheet1”
- Sheet1:
  - Double click on “Sales”, and “Profits”.
  - Then, double click on “Regions”, because there is few options in regions.
- Then we see how much “Sales” and “Profit” done based on the “Regions”.
- Sheet1 (Cntd):
  - Click on the “Shape”, and then on its drop down select one of them and click on “Assign Palette” and ok. You can also increase their sizes by “Size”.

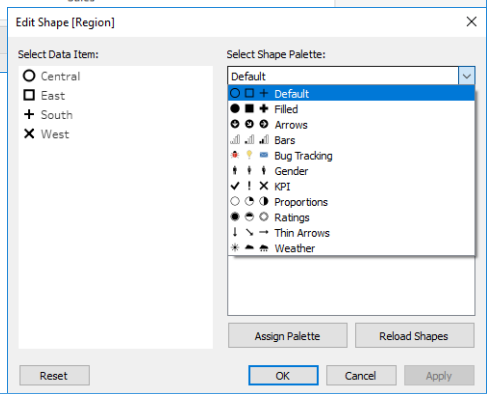
Q4: Which region has the highest “Sales” and “Profit”? How much are they?

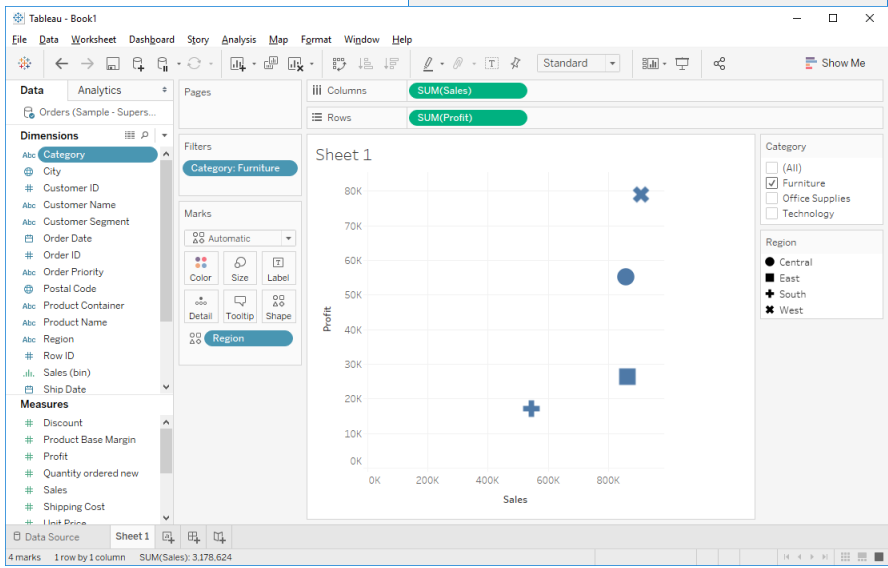
- For creating interactive charts, we have to bring the filter.
- Click on “Category” and its drop down and select “Show Filter”, afterwards on the right hand side you get the filter. Then if you choose only “Furniture”, you get all the options on the Cartesian plane but associated to only “Furniture”.
- You can infer that for “Office Supplies” the profit done by “Central” region was the highest.

Q5: For “Furniture”, which region has the highest profit and how much? Provide a snapshot.

- This quick analysis can be done by interactive scatter charts.
- Clean up the screen

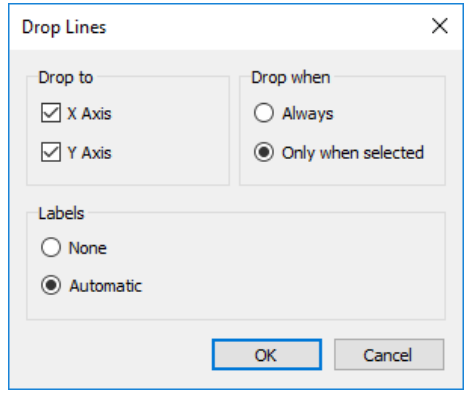


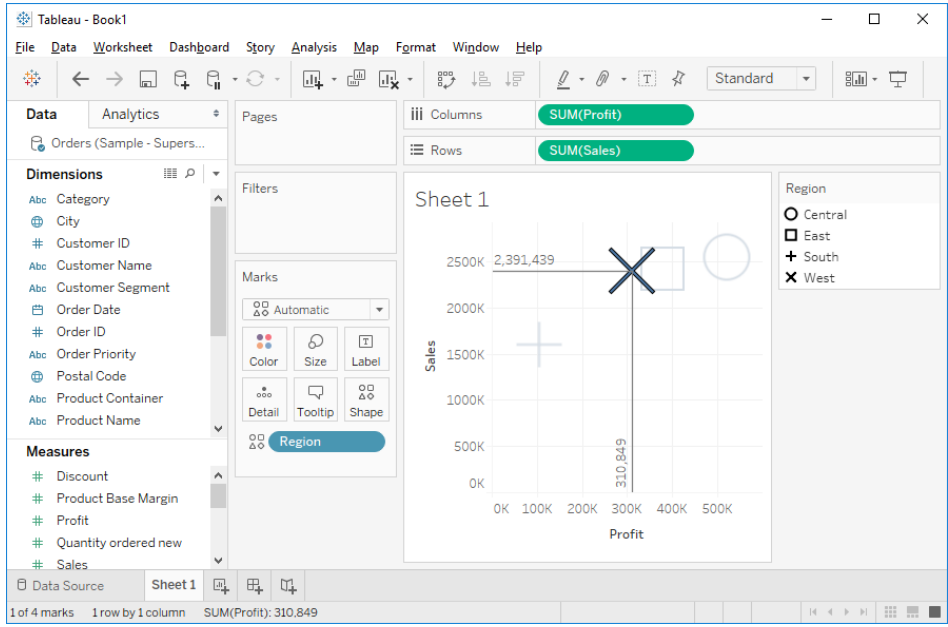




Scatter chart- Drop lines

- Sometimes, it is difficult to understand how much is the exact “Sales” done, or how much is the exact “Profit” done, so we have to use scatter chart with dropped lines.
- Sheet2:
  - Double click on “Sales”, “Profit” and “Region”
  - Now we want to understand the exact “Sales” and “Profit” done.
  - Right click on a point and select “Drop Lines” and “Show Drop Lines”, whenever you select any point on the scatter chart, it shows you a drop line and you can exactly figure out how much it is.
  - If you want numbering at each line, right click on one point and select “Drop Lines” and “Edit Drop Lines” and select “Automatic” on “Labels”. Then you get labels for scattered points.
  - Sometimes, you want only one of “Sales” or “Profit”, not both altogether. In such cases, right click on one point and select “Drop Lines” and “Edit Drop Lines” and choose only one option from “X Axis” or “Y Axis”. Uncheck “Y Axis” box and click ok. So if you select one scattered point, it gives you only one drop line.
  - If you want drop line shown always and for everything, right click on one point and select “Drop Lines” and “Edit Drop Lines” and select “Always” in “Drop When” part. Afterwards, for all options the Y drop lines are shown always.
  - Clean up the screen.

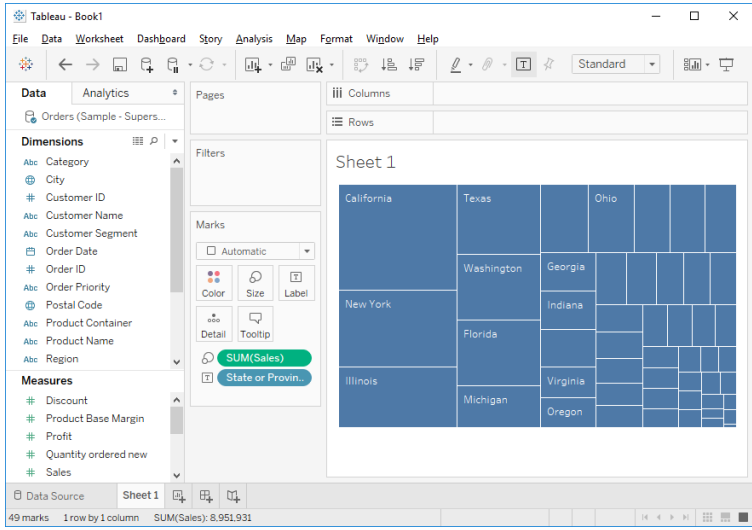


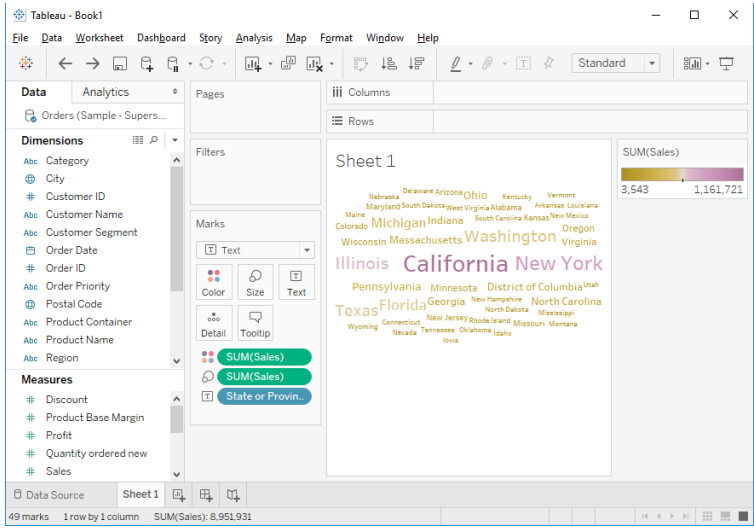


Word maps

Create a Word map

- If you want to represent data (sales) done by any location (country, city, state), the word maps are highly recommended other than bar graphs.
- Different sizes and colors are available based on the “Sales” have done, to create a word map in less than 30 seconds.
- If you want any textual representation, drag that field to the “Text” spot. E.g. pick up the “City” or “State and Province” and drag it in “Text” spot. Then, a document is appeared on the canvas.
- Sheet3:
  - Drag “State and Province” and drop it in “Text” spot
  - Pick up the “Sales” and drop it is “size” spot. Then based on the “Sales”, the size of “State or Province” expands.
  - If you want the word map chart, click on “Automatic” drop down and select “Text”.
  - If you want to display “Sales” based on the colors, drop “Sales” in “Color” part.
  - If you want to change the color palette, select “Color” in “Marks” and click on the “Edit Colors” and select “Gold-Purple Diverging” for instance.
- Sheet4:
  - For creating a Word map in 30 seconds:
  - Drop “Sate and Province” in “Text”
  - Drop “Sales” in “Size” and “Color”
  - Select “Text” from “Automatic”
  - Right click on “Order Priority” and select “Show filter” to create an interactive filter, and keep only “Medium”. Afterwards, few countries based on the “Medium” order priority will be displayed.

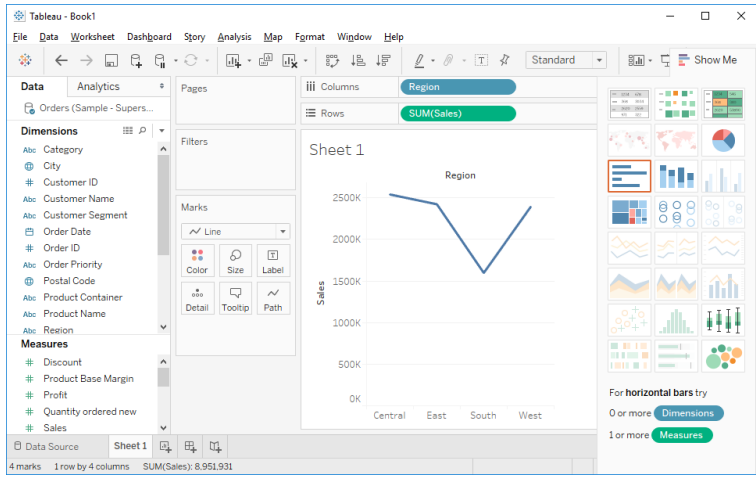


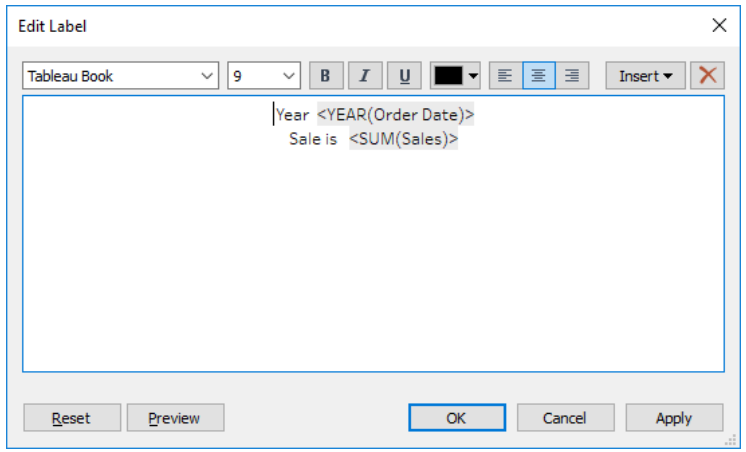


Q6: Provide a snapshot of sheet4. How much is the amount of “sales” done in California with “Medium” order priority?

Line chart

Line chart





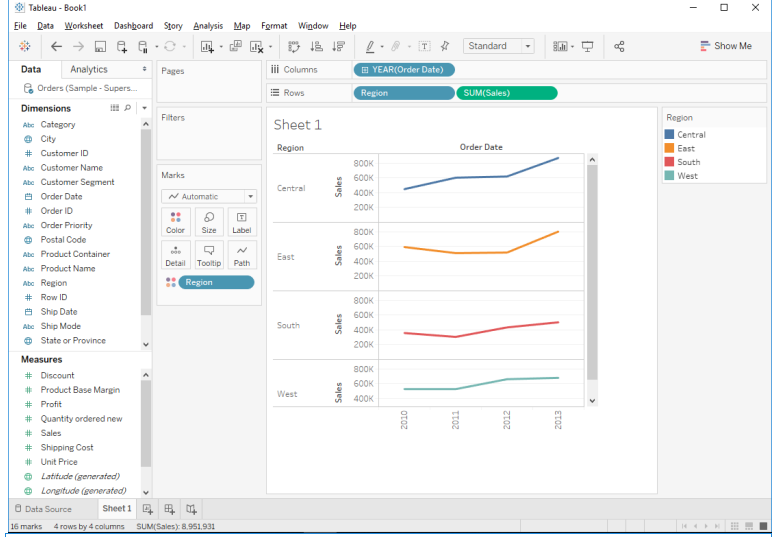
- Whenever you take any fields within “Dimensions” or “Measures”, the visualization is created based on Tableau Visualization Query Language that chooses the best suitable chart. If you want to customize your chart, you can select the best chart.
- Sheet5:
  - “Region” in “Column”
  - “Sales” in “Rows”
  - Bar chart as the best chart that has been automatically created because “Automatic” is shown in “Marks”
  - Change “Automatic” to “Line”, by this way you customize your own chart
- Click on “Show Me”, there are two “Line” charts, one is “continuous” and the other is “discrete”, which need “1 date” compulsory “0 or more Dimensions” and “1 or more Measures”.
- Sheet6:
  - Double click on “Sales” and “Order Date”
  - The automatic “Line” chart is created
  - Drop “Category” in “Color”. Based on each “Category” the line chart is created, which seems that “Technology” had the greatest “Sales” over years. Plus, the highest “Sales” was in 2013.
  - Drop “Order Date” in the “Label”
  - Drop “Sales” into the “Label” to understand how much exact “Sales” done in each year
  - Click on the “Label” and “Text” and add “Year” before the first line and add “Sale is” before the second line to get more meaningful labels on your line chart.

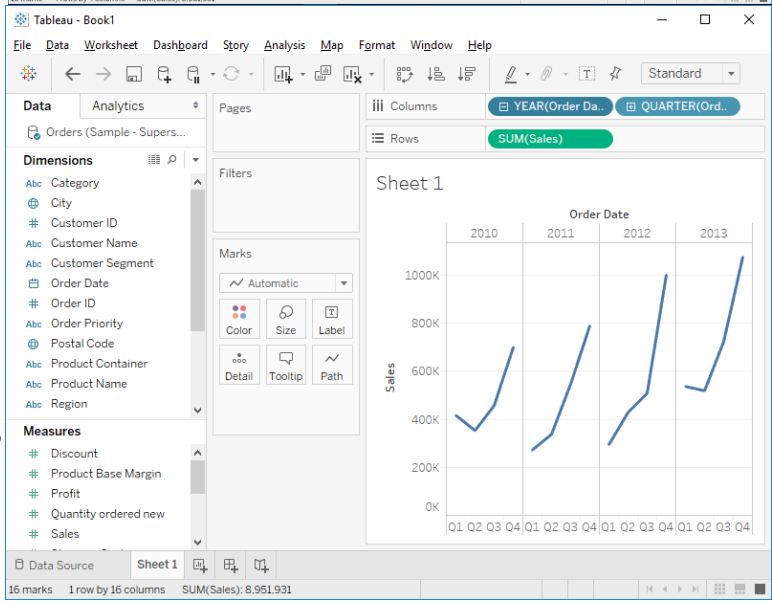
Q7: Provide a snapshot from the jpg image of sheet6 with meaningful title.

Date Line chart

- Sheet7:
  - Double click on “Sales” and “Order Date”
  - Drop “Region” in “Color” and “Column”
  - The sales done based on each region across years are shown
  - Move “Region” to the “Rows”
  - See the difference between the bar and line charts created

Q8: Provide two snapshots from two jpg images of sheet7, one shows bar chart, the other shows line chart. Make sure that they have meaningful titles.



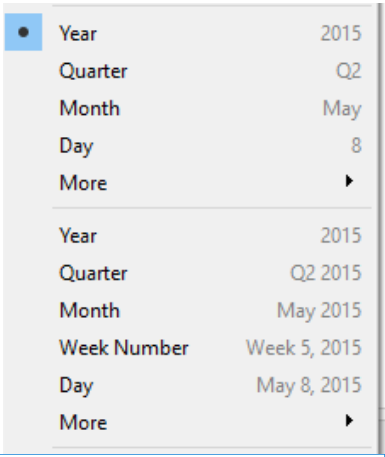


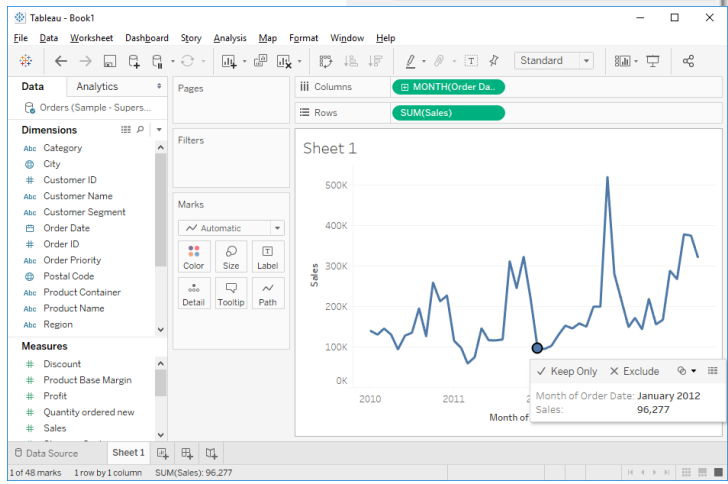
Continuous vs Discrete

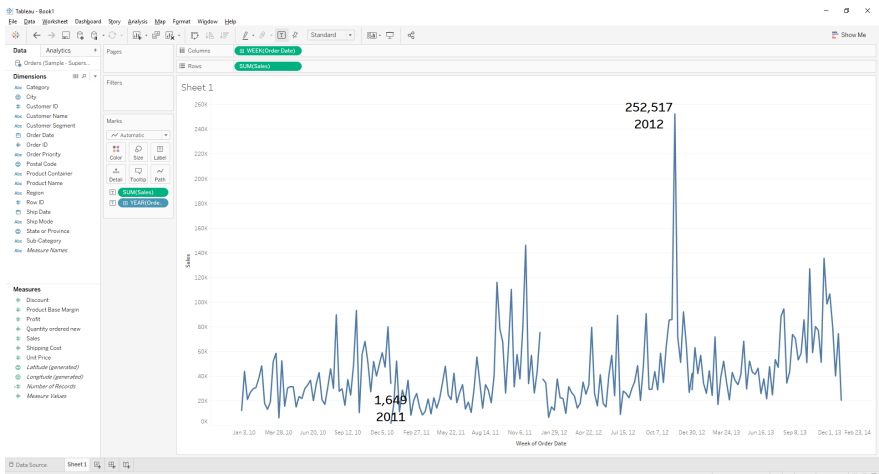
- Coloring of symbols are associated with data types; green fields are continuous, blues fields are discrete.
- There is an exception. “Order Date” can be either continuous and discrete.
- Sheet8:
  - Double click on “Sales” and “Order Date”
  - Drill down the “Order Date”, then different sections are created. This is a discrete chart, which means there is no link between the first and the last one.

Continuous vs Discrete

- Sheet9:
  - Double click on “Sales” and “Order Date”
  - Click on the “Order Date” drop down. The top options are for “Discrete” and the bottom options are for “Continuous”.
  - Click on the “Year” in “Continuous” and see how the graph changes slightly, and the color of “Year” changes to green.
  - Now click on the plus next to “Year” to get the “Month”. Afterwards, the continuous chart is created. Move your cursor on each vertices and see how “Month” of different “Year” s link to each other.
- Tableau is very smart in separating “Continuous” and “Discrete” fields and store them in “Dimensions” and “Measures”, while “Date” field is an exception. It can be both “Continuous” and “Discrete”.
- So, if you are a manager of a company and you want to figure out which year, which month and which day, the highest “Sales” done. Create the following chart less than 30 seconds.
- Sheet10:
  - Double click on “Sales” and “Order Date”
  - Change the “Order Date” to continuous and expand it to “Weeks”
  - Drop “Sales” in “Label”
  - Drop “Order Date” in “Label”
  - Click on “Label” and select “Min/Max” to get your answer



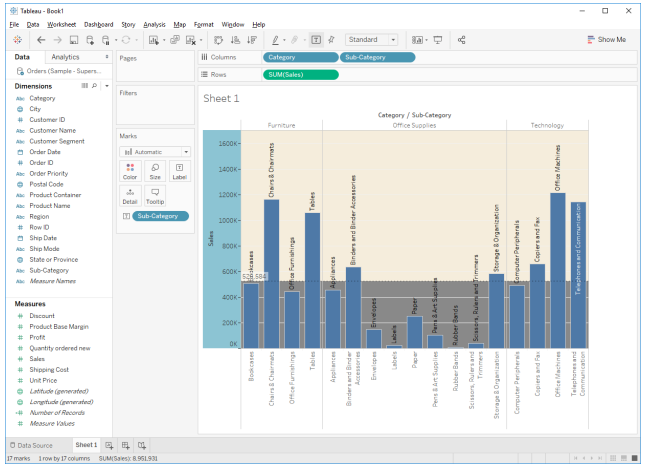


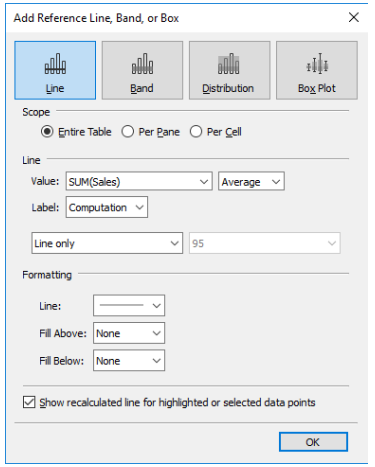


Reference Line

Reference line

- Reference lines are basically used to compare values, which are greater then and less than. For example, creating categories containing sales which are above average and below average. This can be done by references.
- There are various types of references
  - reference line
  - reference band
  - reference distribution
  - reference box plot



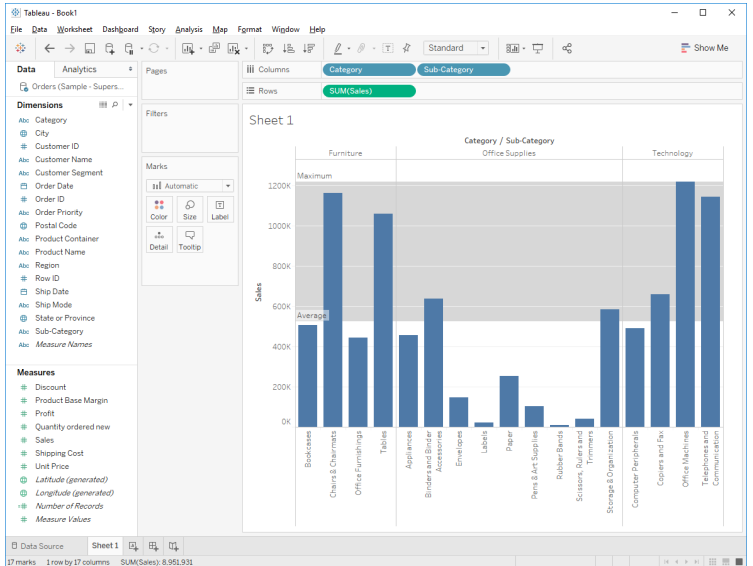


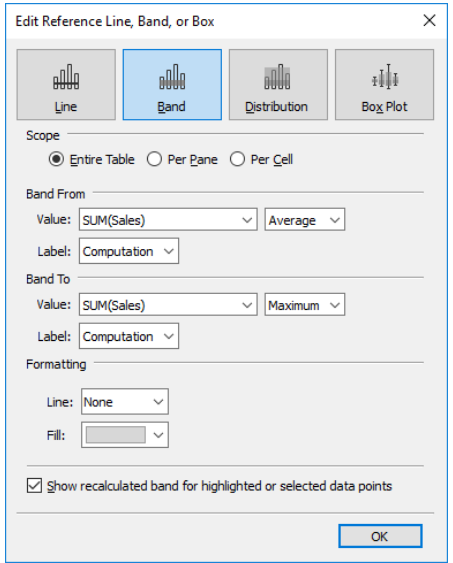
- Sheet11:
  - Drop “Category” and “Sub-Category” into “Columns”
  - Drop “Sales” in “Rows”
  - Suppose that you want to find out “Sales” below and above average
  - Right click on the vertical axis next to “Sales”, click on “add Reference Line”
  - Choose “Line” and “Entire Table”, afterwards the entire table shows a line. Keep “SUM(Sales)” and “Average”.
  - Then, the “Category” and “Sub-Category” which had “Sales” done below and above average can be seen. Drop “Sub-Category” in “Label” part.
  - If you want to see the exact value of the reference line, Right click on the vertical axis next to “Sales”, click on “Edit Reference Line”, change the “Label” into “Value”. Afterwards, the exact value of average is shown on the chart.
  - In “Edit Reference Line” window, change the options in “Formatting” and change “Line” format and “Fill below” format. So we can figure out which Category and Sub-Category are doing better.
  - In “Edit Reference Line” window“ select “Per pane” creates reference lines in each pane separately. So the manager can push up the Categories that are not doing well.

Q9: Provide a jpg images from sheet11. Which sub categories in “Technology” had “Sales” less than its average?

Reference Distribution Line

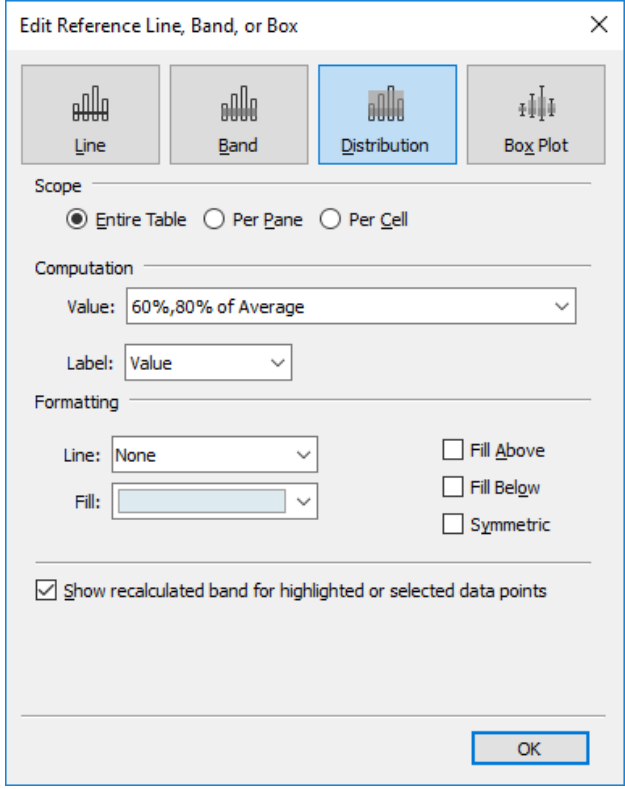
- Sheet12:
  - Drop “Category”, “Sub-Category” in “Columns”
  - Drop “Sales” in “Rows”
  - Right click on the vertical axis next to “Sales”, click on “add Reference Line”
  - Select “Band”, “Entire Table”
  - When you want to create a band, you have to define “Band From” and “Band To”, where the band starts and ends respectively. Choose “Average” for “Band From” and “Maximum” for “Band To”.
  - Then the band is created, which colors area from “Average” to “Maximum”
  - You can change it to “Minimum” and “Average” and band’s color in “Formatting”
  - If you want to see the exact values, change both “Label” s into “Value”
  - If you want to modify options of reference line, Right click on the vertical axis next to “Sales”, click on “Edit Reference Line”





Reference band

- Sheet13:
  - Drop “Category”, “Sub-Category” in “Columns”
  - Drop “Sales” in “Rows”
  - Right click on the vertical axis next to “Sales”, click on “add Reference Line”
  - Select “Distribution”, “Entire Table”. You can divide the “Sales” into percentage wise.
  - By default, area between 60% and 80% of Average is colored. If you want to change these default values, click on drop down and type other values such as “50,100”, select “Value” for “Label”.
  - You can also write more than two numbers such as “60,70,80” and see how the distribution bands created
  - Then the created distribution band and the upper, middle and lower values are shown.
  - There are other options such as “Percentile”, “Quantile”, and “Standard Deviation” that have been offered.



Q10: Provide a snapshot from jpg image of sheet13, with three values in distribution (60,70,80).

